

An open Python emulator of the OBR macroeconomic model for UK fiscal-reform scoring^{*}

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Abstract

I describe an open-source Python implementation of the Office for Budget Responsibility’s macroeconomic model of the UK economy, built for near-term fiscal-reform scoring in the PolicyEngine Macro suite. The emulator transpiles the OBR’s 372 published EViews equations into Python, solves them by Gauss–Seidel iteration with a demand-side closure swap, and anchors the baseline to the OBR’s November 2025 *Economic and Fiscal Outlook* using the OBR’s own add-factor device. In anchored mode the model reproduces the published EFO path with a GDP MAPE of 0.18 per cent and consumption of 0.29 per cent over 2025Q1–2027Q4, hard-gated in continuous integration. The free-running fit is deliberately reported without flattery: of 21 headline variables, 10 are passthroughs and only 4 of the 11 actually computed land within band, with company profits 54.6 per cent off—a gap traced to unpublished OBR calibration constants rather than live bugs. A static-costing bridge injects PolicyEngine microsimulation costings into the model as a household-disposable-income shock, delivering the OBR’s own “static costing in, second-round effects out” workflow. A worked example—raising the basic rate of income tax by 1p—yields a static costing of £6.46 billion in 2026 rising to £7.38 billion by 2030, within the range of HMRC’s published ready reckoner, with second-round GDP effects of -0.02 to -0.03 per cent.

Keywords: macroeconomic models, fiscal policy, policy costings, OBR, microsimulation, open source

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1 Introduction

When the UK government changes a tax or benefit, two questions follow immediately. The first is static: who pays what the morning after? The second is dynamic: how do consumption, investment, incomes, and prices adjust over the following years, and what does that adjustment do to the revenue estimate? The Office for Budget Responsibility (OBR) answers the second question with a large-scale macroeconometric model in the Treasury/Cowles-Commission tradition—a model whose equation listing and variable definitions are published, but which is operated in proprietary EViews software, on a databank the OBR does not fully publish, and with judgement imposed through unpublished residual adjustments (Office for Budget Responsibility, 2013, 2025b).

This paper describes an open-source Python emulator of that model, built as the near-term fiscal-scoring member of the *PolicyEngine Macro* suite.¹ The emulator takes the OBR’s published model code of 15 October 2025—372 equations—transpiles it mechanically into Python, initialises it from the November 2025 *Economic and Fiscal Outlook* (EFO) detailed forecast tables (Office for Budget Responsibility, 2025a) and a 348-series snapshot of Office for National Statistics data, and solves it by Gauss–Seidel iteration. Policy analysis proceeds the way the OBR’s own does: the baseline is anchored to the published forecast through add-factors, a shock is applied to an exogenous lever (or injected through a microsimulation bridge), and the reform effect is read as the deviation between the shocked and baseline solutions.

The contribution is threefold. First, *replication as infrastructure*: to my knowledge this is the first publicly runnable implementation of the OBR’s macroeconomic model. The OBR publishes the equation listing and a briefing paper describing the model (Office for Budget Responsibility, 2013) but no official working paper accompanies the model itself, and the model cannot be run without EViews and the OBR’s internal databank. The emulator makes the official forecaster’s transmission mechanisms—the marginal propensities, error-correction speeds, and accounting identities that shape UK fiscal scoring—inspectable and executable by anyone with Python.

Second, *integration with microsimulation*: a static-costing bridge connects PolicyEngine’s UK tax–benefit microsimulation (PolicyEngine, 2025) to the macro model, reproducing the OBR’s own policy-costing workflow (Office for Budget Responsibility, 2014) in which a static costing from the revenue departments enters the macroeconomic forecast and second-round demand effects come out. A reform expressed as PolicyEngine parameters is costed at household resolution, converted to a shock on nominal household disposable income, and propagated through the model’s consumption and expenditure blocks.

Third, *honest validation*: the project distinguishes sharply between what the model reproduces by construction and what it genuinely forecasts. The anchored baseline matches the EFO to within 0.2–0.3 per cent by design and this invariant is hard-gated in continuous integration; the free-running model, scored without the add-factors and de-seeded from the EFO path, is documented as weak—only 4 of 11 computed headline variables land within band—with each residual error traced either to a fixed bug or to a documented “floor” imposed by constants the OBR does not publish. Section 6 places the model on the suite’s *verification gradient*, which grades each model class by how far its outputs can be audited against ground truth.

The rest of the paper proceeds as follows. Section 2 sets out the model’s equation structure,

¹<https://macromod.vercel.app/obr/>; source at <https://github.com/PolicyEngine/obr-macroeconomic-model>. PolicyEngine Macro is a suite of open macroeconomic simulation models—overlapping-generations, macroeconometric, and structural-VAR classes—that score the same PolicyEngine reform objects and report comparable real-world aggregates (PolicyEngine, 2025).

including the key behavioural equations in their published form, the error-correction and add-factor architecture, and the anchoring mechanism. Section 3 describes the implementation: transpiler, solver, closure swap, microsimulation bridge, and hosted interfaces. Section 4 presents calibration and validation against the November 2025 EFO. Section 5 reports a worked reform—1p on the basic rate of income tax—and compares it with HMRC’s published ready reckoner (HM Revenue and Customs, 2025). Section 6 discusses limitations, and Section 7 concludes.

2 The model

The OBR’s macroeconomic model descends from the Treasury model first built in 1970 and has been jointly maintained by the OBR and HM Treasury since 2010 (Office for Budget Responsibility, 2013). It is a quarterly, large-scale macroeconomic model in the estimated-structural tradition of Klein (1950) and Fair (2004): a small number of estimated behavioural equations embedded in a dense web of national-accounts identities, with expectations largely implicit and adjustment dynamics carried by error-correction terms (Davidson et al., 1978; Engle and Granger, 1987). It sits in the lineage of the large UK policy models reviewed by Wallis et al. (1987), and is the UK institutional counterpart of semi-structural policy models such as the Federal Reserve’s FRB/US (Brayton and Tinsley, 1996). The 15 October 2025 model listing used here contains 372 equations over 636 variable codes. Of these, roughly 29 equations are econometrically estimated, 216 are accounting identities, and 138 are calibrated or technical relationships; 253 variables are exogenous inputs—the fiscal levers, market assumptions, world demand, and demographic paths on which the OBR conditions its forecast—and 383 are solved by the model.

2.1 The expenditure core

Real GDP at market prices ($GDPM$) is tied to its expenditure components by the identity

$$GDPM = CGG + CONS + IF + DINV + VAL + X - M + SDE, \quad (1)$$

where CGG is government consumption, $CONS$ private consumption, IF total fixed investment, $DINV$ inventory change, VAL valuables, X and M exports and imports, and SDE the statistical discrepancy. In the published listing this identity is arranged with inventories as the residual; the emulator’s closure swap (Section 3) re-arranges it so that output responds to demand shocks.

2.2 Consumption

The centrepiece behavioural equation is an error-correction consumption function in the tradition of Davidson et al. (1978). As published (coefficients verbatim from the OBR listing):

$$\begin{aligned} \Delta \ln CONS = & 0.2646 + 0.1030 \Delta \ln RHHDI - 0.0084 \Delta LFSUR \\ & + 0.1269 \Delta \ln \left(\frac{1000 GPW}{PCE/100} \right) - 0.0004 \Delta [R_{-1} - \pi_4^{PCE}] \\ & - 0.1251 \left[\ln CONS_{-1} - 0.4393 \ln RHHDI_{-1} \right. \\ & \quad \left. - 0.1059 \ln \left(\frac{1000 GPW_{-1}}{PCE_{-1}/100} \right) - 0.2216 \ln \left(\frac{NFWPE_{-1}}{PCE_{-1}/100} \right) \right], \end{aligned} \quad (2)$$

where $RHHDI$ is real household disposable income, $LFSUR$ the LFS unemployment rate, GPW gross housing wealth, $NFWPE$ net financial wealth of the personal sector, PCE the consumption deflator, R Bank Rate, and π_4^{PCE} annual consumption-deflator inflation. The long-run solution ties consumption to income, housing wealth, and financial wealth with an adjustment speed of about 12.5 per cent per quarter; the short run responds to income growth, unemployment changes, wealth growth, and the real interest rate. A companion equation splits off durables ($CDUR$) with relative-price, income, and user-cost terms.

2.3 Investment

Total investment is built from its components,

$$IBUS = IF - GGI - PCIH - PCLEB - IH - IPRL, \quad (3)$$

with business investment behaviour specified as an error-correction accelerator on market-sector output with a user-cost channel:

$$\begin{aligned} \Delta \ln IBUSX = & 0.1992 \Delta \ln IBUSX_{-3} + 1.0057 \Delta \ln MSGVA_{-1} - 0.0012 CBIUD \\ & - 0.0418 [\ln IBUSX_{-1} - \ln(1000 KMSXH_{-2}) + KGAP_{-2}] - 0.0884, \end{aligned} \quad (4)$$

where $MSGVA$ is market-sector gross value added, $CBIUD$ a CBI capacity-utilisation survey balance, $KMSXH$ the market-sector capital stock, and $KGAP$ the gap between actual and desired capital—the term through which the corporation-tax rate ($TCPRO$), by raising the user cost of capital, lowers desired capital and hence investment. In the October 2025 listing this behavioural line ships commented out (the OBR runs investment off its databank with judgement); the emulator’s corporation-tax experiments therefore operate the channel through an explicit investment closure with the cost-of-capital relationship restored and stabilised (Section 3).

2.4 Incomes

Household disposable income is a pure accounting identity over the sector accounts:

$$\begin{aligned} HHDI = & MI + FYEMP - EMPSC - EESC - TYWHH + NMTRHH + SBHH \\ & + (PIRHH - PIPHH + FSMADJ) - HHSB + HHISC + (EECOMPC - EECOMPD) + OS \end{aligned} \quad (5)$$

summing mixed income, employee compensation less social contributions, taxes on income ($TYWHH$), net transfers and benefits, net property income, and households’ operating surplus ($OSHH$, mostly imputed rent). Real income deflates by the consumption deflator, $RHHDI = 100 HHDI/PCE$, and feeds equation (2). This chain— $HHDI \rightarrow RHHDI \rightarrow CONS \rightarrow GDPM$ —is the transmission path used by the microsimulation bridge. On the other side of the factor-income split, company profits are the residual of the operating-surplus account:

$$FYCPR = OS - OSHH - OSGG - OSPC - RENTCO + SA - FISIMPS. \quad (6)$$

The wage bill aggregates employment, average earnings, and public-sector pay:

$$WFP = ADJW \cdot PSAVEI \cdot (EMS - ESLFS) + \frac{52}{4000} CGWADJ \cdot ERCG \cdot ECG + \frac{52}{4000} LAWADJ \cdot ERLA \cdot ELA, \quad (7)$$

with private-sector average earnings ($PSAVEI$) determined by an estimated error-correction equation in market-sector prices, productivity, unemployment, and the wedge between consumer

and producer prices. Employment is the identity $ETLFS = 1000 (HWA/AVH)$, total hours over average hours—both exogenous inputs in this configuration, a point that matters for the honest scorecard in Section 4.

2.5 Prices

The retail-price block builds the RPI from mortgage-interest and other components, $PR = I7 [(1-W_4) PRXMIP/I9 + W_4 PRMIP/I4]$, with annual RPI inflation $RPI = 100 (PR/PR_{-4} - 1)$; the consumption deflator moves with the CPI through $PCE/PCE_{-4} = CPI/CPI_{-4}$. The CPI itself is exogenous in this configuration (the OBR imposes its inflation judgement), so the deflator path is conditioned on the EFO inflation forecast.

2.6 Fiscal block

Government receipts and spending aggregate from their components—for example income-tax receipts $INCTAXG = TYEM + TSEOP + TCINV - INCTAC + CTC - NPISHTC$ —up to public sector net borrowing, $PSNB = -PSCB + PSNI$ (the current deficit plus net investment). The fiscal levers a reform can pull directly include government consumption CGG , nominal central-government investment $CGIPS$, the corporation-tax rate $TCPRO$, and the market assumptions (Bank Rate R , gilt yields, the exchange rate).

2.7 Add-factors and the anchoring mechanism

Every estimated equation in the listing carries an add-factor declaration (EViews `@ADD`), the residual-adjustment device through which the OBR imposes judgement on the pure model solution ([Office for Budget Responsibility, 2013](#)): for a behavioural equation $y_t = f(x_t) + a_t$, the add-factor a_t shifts the equation’s intercept period by period. The emulator uses the same device for two distinct purposes.

Anchoring. For the baseline, add-factors are computed at every quarter as exactly the residuals that reconcile each behavioural equation to the published EFO path, so the anchored baseline reproduces the OBR’s published aggregates *by construction*. Two published income aggregates ($HHDI$, $RHHDI$) are computed by level identities that receive no add-factor; because their unpublished inputs drift, they are instead held exogenously at their EFO values in the anchored solve—closing a leak that otherwise pulls consumption (through the $\Delta \ln RHHDI$ term) about 2 per cent off the EFO. This is anchoring to ground truth, not snapshotting a model error: the shock and baseline runs share the identical anchored structure, so reform deltas isolate the policy effect.

Forecasting. For genuine forecasts the emulator implements the OBR’s own held-add-factor method: residuals are fitted over a recent base window (2024Q1–2025Q4), held constant, and the model is projected forward eight quarters. The forecast is the model’s structural dynamics anchored by the frozen judgement terms—a real forecast rather than a replay of the input.

Exogenous variables. The 253 exogenous inputs comprise the fiscal levers above, monetary and financial-market assumptions, world demand and trade prices, demographics, and the technical adjustment factors. Where the EFO does not publish an input the model needs, the emulator draws on a vendored ONS snapshot: of 261 endogenous variables referenced by the listing but absent from the EFO tables, 199 had identifiable ONS series codes, yielding a 348-series databank that gives the circular fiscal and financial sub-blocks real balancing data.

3 Method and implementation

The emulator is a pure-Python package (`obr_macro`, Python 3.11–3.13, NumPy/pandas) that ships the OBR equation listing, the November 2025 EFO detailed tables, and the ONS input snapshot as package data, so `pip install` from GitHub is sufficient to run it. The pipeline has five stages.

1. Transpile. A deterministic transpiler rewrites each EViews line into an equivalent Python expression: `dlog(X)` and `d(X)` become log- and first-differences with balanced-parenthesis scanning, lags `X(-n)` become indexed look-backs, `@elem(VAR, "1970Q1")` becomes a base-period constant lookup, `@recode(@date=...)` becomes a period dummy (with EViews month-coded quarters normalised), `@trend` becomes a time index, and `^` becomes `**`. Each equation is classified (identity, error-correction, growth-rate, simultaneous) and compiled once; the dashboard's Equations tab displays every published EViews line beside its transpiled Python for line-by-line audit.

2. Initialise. History and forecast values load from the EFO tables and the ONS snapshot from 2016Q1; any variable an equation references but the data do not supply is created and seeded. Left-hand sides that arrive as transformations ($\Delta \ln y$, Δy , ratios) are solved analytically for the level of y .

3. Solve. Each quarter's 372 interdependent equations are solved simultaneously by Gauss–Seidel iteration (Fair, 2004): cycle through the equations in order, update each left-hand variable in place, and repeat until the maximum change falls below tolerance (with damping and stall detection). Quarters are solved sequentially forward over the horizon; scenario solves run 2025Q1–2027Q4 and baseline solves extend to 2031Q1.

4. Closure swap. The published listing fixes GDP and lets identities distribute it; for policy shocks the emulator swaps the closure, re-arranging identity (1) so that *GDPM* becomes endogenous and inventories *DINV* become the residual. Output then responds to demand. Under this demand closure the impact multiplier on a government-consumption shock is approximately one by construction (£1.25bn per quarter of *CGG* raises quarterly GDP by about £1.25bn on impact). For corporation-tax experiments an optional *investment closure* restores the user-cost channel: *TCPRO* moves the cost of capital, desired capital, and business investment via equation (4), with the market-sector supply feedback stabilised by tracking-pass add-factors to prevent an explosive output–investment accelerator loop (an eigenvalue-above-one interaction that was diagnosed, bounded, and locked with sign- and magnitude-invariant tests).

5. Shock and compare. `run_reform(name, var, shock, periods)` solves the baseline and the shocked path with identical structure—add-factors on in both—and reports per-quarter deviations (ΔGDP in £bn, per cent of GDP, Δ consumption, Δ investment). Shocks may be scalar (£m per quarter, or decimal rate changes) or explicit quarterly paths.

3.1 The static-costing bridge from PolicyEngine microsimulation

The bridge reproduces the OBR's policy-costing workflow (Office for Budget Responsibility, 2014): static costing in, second-round effects out. A reform is expressed as PolicyEngine

parameters—the same `{parameter_path: value}` dictionary used across the suite, e.g. `{"gov.hmrc.income_tax": 0.21}`—and processed in three steps:

1. *Microsimulation costing.* PolicyEngine’s UK model (PolicyEngine, 2025) runs the reform against the enhanced Family Resources Survey microdata for each year of the scoring window, returning the annual budgetary impact in £bn together with household-level detail (winners, losers, net-income change).
2. *Translation.* Annual costings convert to a quarterly additive shock on nominal household disposable income (*HHDI*, £m per quarter): sign-flipped (revenue raised means households keep less), scaled by 1000/4, and applied flat within each year—a deliberately crude and declared interpolation, since the microsimulation produces annual numbers.
3. *Macro propagation.* `run_reform(var="HHDI", shock=path)` propagates the shock through the OBR’s own demand-side incidence channel, $HHDI \rightarrow RHHDI \rightarrow CONS \rightarrow GDPM$ (equations (5) and (2)), with *HHDI* exogenised identically in baseline and shocked runs so the delta isolates the reform. The output is the second-round demand effect.

Two boundaries are enforced rather than fudged: the economy’s feedback onto disposable income itself (income $\rightarrow$ tax $\rightarrow$ income) is not recycled, and corporation-tax reforms are refused by the bridge—corporation tax is not household-borne in the microsimulation—with a pointer to the direct *TCPRO* lever. Supply-side channels (participation, saving, capital deepening) are deliberately left to the suite’s overlapping-generations member; that division of labour is the point of a multi-model suite.

3.2 Hosted surfaces

The emulator is drivable from AI workflows and the command line through the suite’s integration layer (`pe-macro`): a CLI and a Model Context Protocol (MCP) server exposing tools including `score_reform` (a PolicyEngine reform through a chosen model, with a common `ScoreResult` schema across model classes), `obr_shock` (raw variable shocks), forecasting, and household calculators. The hosted MCP server deploys automatically to Modal on every merge—merges to the model repository trigger the same redeploy—and a live dashboard (<https://obr-macroeconomic-model.vercel.app>) provides the scenario explorer, a searchable glossary of all 636 variable codes with ONS series, and the full equation listing beside its transpiled Python. Nightly CI validates the live endpoint against the correctness suite.

4 Calibration and validation

All results in this paper are conditioned on the November 2025 EFO vintage (Office for Budget Responsibility, 2025a) and the 15 October 2025 model code. Validation is organised around a principle borrowed from the reliability literature: separate what the model reproduces *by construction* from what it genuinely generates, and gate the former hard while reporting the latter honestly.

4.1 The anchored baseline: a hard gate

With add-factors on (Section 2.7), the anchored baseline must reproduce the OBR’s published aggregates—that is what anchoring means, and any deviation is a broken transmission chain, not

a forecast error. Table 1 reports the anchored fit over the scored horizon 2025Q1–2027Q4. The GDP mean absolute percentage error is 0.18 per cent and consumption 0.29 per cent; continuous integration hard-fails the build if either exceeds 1 per cent, if any published aggregate turns non-finite on the horizon, or if the expenditure identity (1) fails to close within 0.5 per cent of GDP in any quarter. (An earlier version of the anchored solve missed the EFO by 1.93 per cent on GDP because $HHDI/RHHDI$ ran raw and leaked into the consumption equation; anchoring the published income aggregates closed the gap to 0.18 per cent.)

Table 1: Anchored baseline versus the OBR November 2025 EFO, 2025Q1–2027Q4

Invariant	Metric	Value	CI gate
Real GDP ($GDPM$)	MAPE vs EFO	0.18%	hard fail if $\geq 1\%$
Consumption ($CONS$)	MAPE vs EFO	0.29%	hard fail if $\geq 1\%$
Key aggregates finite	NaN/inf on horizon	none	hard fail on any
Expenditure identity	max gap, % of GDP	$< 0.5\%$	hard fail if $\geq 0.5\%$
Spending multiplier sign	$\Delta GDPM$ for $+CGG$	positive	hard fail on sign flip
Corp.-tax investment sign	ΔIF for $\pm TCPRO$	correct, bounded	hard fail on sign/blow-up

Notes: The anchored fit is a by-construction invariant, not a forecast claim: add-factors absorb the model’s tracking error so the baseline reproduces the published EFO path. Structural sign invariants (positive fiscal multiplier; a corporation-tax cut raises, and a rise lowers, investment, with the response bounded) are gated alongside it. Source: `tests/test_model_invariants.py` in the model repository.

4.2 The free-running model: an honest scorecard

The true measure of calibration is the raw model with add-factors off. Two flattery traps are removed before scoring. First, *de-seeding*: the solver’s data frame is pre-filled with the EFO path, so a quarter whose Gauss–Seidel iteration exits on a stall would otherwise sit near the EFO seed; the scorecard re-solves each period after reseeding every live computed variable from the model’s own previous-period value. Second, *passthrough labelling*: a 0.00 per cent error usually does not mean a good forecast—it means the variable was left at the OBR’s published value because its equation is dead or exogenous. Only variables the model actually computes measure calibration, and the employment identity ($ETLFS = HWA/AVH$, both passthrough inputs) is flagged as trivial rather than counted as a behavioural win.

Table 2 presents the scorecard. “Within band” collapses per-metric thresholds—rates within 1.0pp, net balances within 1.5 per cent of GDP (the OBR’s own convention for scoring a difference of two large gross flows), levels within 10 per cent MAPE.

Three residual errors deserve explanation because they are *floors*, not bugs. The company-profits error (54.6 per cent) traces to households’ operating surplus: the published equation $OSHH = 12874 + 0.85IROO - DIPHH^{mf}$ evaluates to roughly £38bn against the OBR databank’s roughly £77bn, and the gap lives in the additive base constant and the vintage of the unpublished rent index feeding $IROO$. Because $HHDI$ adds $OSHH$ (income runs ~ 15 per cent low) while $FYCPR$ subtracts it (profits are inflated), one unpublished constant drives a large share of the error in three scored variables at once. Re-tuning the constant to hit the target would be fitting to the answer, so it is documented and gated against regression instead. Similarly, RPI inflation decays because the unpublished $I7/PR$ index history runs out around 2026Q2, and the current-account error is bounded by overseas rate-of-return normalisers the

Table 2: Free-running (raw) model versus the OBR November 2025 EFO, 2025Q1–2027Q4

Variable	Metric	Error (3yr)	Status
Real GDP	MAPE	5.70%	computed — fair
Consumption	MAPE	9.45%	computed — fair
Trade balance	% of GDP	0.62%	computed — fair
Employment	MAPE	0.00%	computed — <i>trivial identity</i>
Unemployment rate	pp	1.01	computed — poor
RPI inflation	pp	2.03	computed — poor
Household income	MAPE	14.76%	computed — poor
Real household income	MAPE	14.48%	computed — poor
Business investment	MAPE	15.48%	computed — poor
Current account	% of GDP	2.76%	computed — poor
Company profits	MAPE	54.57%	computed — off
Investment, exports, imports, employees, avg. earnings, CPI, CPI inflation, GDP deflator, wages, compensation		0.00%	passthrough (held at OBR value)

Notes: De-seeded raw solve, add-factors off. Of 21 headline variables, 11 are computed and 10 are passthroughs; of the 11 computed, 4 land within band (36%)—real GDP, consumption, the trade balance, and the trivial employment identity. Bands: rates <1.0pp, net balances <1.5% of GDP, levels <10% MAPE. Source: `docs/calibration.scorecard.md`, generated by `obr_macro/calibration.score.py`.

OBR does not publish. These figures are far better than early development versions (business investment 318 per cent, household income ~932 per cent, company profits ~375 per cent), which reflected genuine bugs in the incomes, investment, and financial-account blocks that were found and fixed; what remains is bounded by unpublished inputs.

4.3 Held-add-factor forecast

Between the two extremes sits the forecasting configuration (Section 2.7): add-factors fitted over 2024Q1–2025Q4, held constant, projected over 2026Q1–2027Q4. Scored against the EFO, real GDP comes in at 2.2 per cent MAPE, consumption 3.6 per cent, unemployment 0.75pp, and the current account 0.9 per cent of GDP—within band—while business investment (~18 per cent), company profits (~47 per cent), and household income (~12 per cent) sit over band: 10 of 16 headline variables are computed and 6 of those land within band. Honesty caveats apply and are documented with the result: the forecast is initialised at the EFO values it is scored against (a neutral jump-off scores worse); the add-factor base window extends into quarters that are themselves OBR forecast rather than outturn, so the held add-factors partly encode “agree with the OBR”; and the ONS input snapshot is current-vintage data merged into an autumn-2025 baseline, so revision slack is absorbed into the add-factors.

5 Results: a worked reform

To illustrate the full pipeline, consider raising the basic rate of income tax by one percentage point (20p to 21p) from April 2026—the canonical illustrative change in HMRC’s ready reckoner. Expressed as a PolicyEngine reform it is a single parameter, `{"gov.hmrc.income_tax_rates.uk[0].rate": 0.21}`, scored through the bridge over a five-year window, 2026–2030.

Static costing. PolicyEngine’s microsimulation puts the direct Exchequer yield at £6.46 billion in 2026, rising with fiscal drag and earnings growth to £7.38 billion by 2030 (Table 3). Household net income falls by a corresponding amount, and losers far outnumber winners—as expected for a broad-based rate rise.

Second-round effects. Converted to a quarterly *HHDI* shock (roughly −£1.6bn to −£1.8bn per quarter) and propagated through the consumption function (2) and the expenditure identity, the reform lowers GDP by 0.02–0.03 per cent relative to baseline over the window: households smooth part of the income loss (the short-run marginal propensity to consume out of income in equation (2) is modest, with the error-correction term pulling consumption toward its long-run income-and-wealth solution at ~12.5 per cent per quarter), so the demand drag is small relative to the mechanical costing. This is the sign and order of magnitude the OBR’s own indirect-effect adjustments deliver for income-tax measures (Office for Budget Responsibility, 2014).

Table 3: Basic rate of income tax +1pp, scored through the static-costing bridge

	2026	2030
Static costing (PolicyEngine microsimulation), £bn/yr	+6.46	+7.38
Quarterly <i>HHDI</i> shock, £bn/qtr	−1.62	−1.85
Second-round GDP effect (OBR emulator), % of GDP	−0.02	−0.03
HMRC ready reckoner, 1p on basic rate, £bn/yr	≈ 6–8 (nearly 8 in 2026–27)	

Notes: Reform `{"gov.hmrc.income_tax_rates.uk[0].rate": 0.21}` from 2026, scored via `obr.score.reform` (Section 3.1): PolicyEngine annual costings on enhanced FRS microdata, converted to a flat-within-year quarterly shock on nominal household disposable income, propagated through the anchored OBR emulator under the demand closure. The GDP effect is the deviation of *GDPM* from the anchored baseline. HMRC comparison: HM Revenue and Customs (2025).

Comparison with HMRC’s ready reckoner. HMRC’s *Direct effects of illustrative tax changes* bulletin—the official ready reckoner, produced on the OBR-certified costing basis—puts a 1p change in the basic rate at close to £8 billion a year in 2026–27 (HM Revenue and Customs, 2025), and published vintages over recent years span roughly £6–8 billion per 1p. The PolicyEngine static costing of £6.46–7.38 billion sits within that range, toward its lower end; differences of this size are expected across costings that differ in data vintage (survey microdata versus HMRC administrative Survey of Personal Incomes), baseline policy assumptions (threshold-freeze paths), and behavioural adjustments (HMRC’s figures embed taxable-income elasticities for higher incomes). Under the OBR’s policy-costing conventions (Office for Budget Responsibility, 2014), the ready-reckoner figure corresponds to the post-behavioural *direct* effect, while the economy-wide feedback is applied at the forecast level—exactly the decomposition the bridge reproduces: the microsimulation supplies the direct effect, the macro model the indirect one.

Direct macro levers. Reforms outside the household tax–benefit space use the exogenous levers directly. A £5bn/yr government-consumption shock (*CGG* +£1.25bn per quarter) raises GDP approximately one-for-one on impact under the demand closure; a 5pp corporation-tax cut with the investment closure raises business investment through the user cost of capital, with sign and boundedness locked by the CI invariants of Table 1.

6 Limitations

The PolicyEngine Macro suite grades its members on a *verification gradient*: how far each model’s outputs can be audited against ground truth should temper the weight placed on its answers. The OBR emulator is a *replication with a published anchor*—auditable against the EFO tables—which is a stronger epistemic position than a calibrated counterfactual, but the audit itself reveals clear limits. In the spirit of reporting them rather than rounding them away:

1. **The anchored fit is by construction, not skill.** The 0.18/0.29 per cent reproduction of the EFO (Table 1) demonstrates that the transmission chains are wired correctly; it says nothing about forecasting ability. All forecasting claims rest on the much weaker free-running and held-add-factor scorecards.
2. **The free-running model is weak, and says so.** Only 4 of 11 computed headline variables land within band; 10 of 21 are passthroughs held at the OBR value (Table 2). The suite’s documentation describes the free-running fit as “deliberately report-only, and weak,” and this paper endorses that description.
3. **Unpublished inputs impose hard floors.** The OBR does not publish its full databank: the *OSHH* base constant (behind the 54.6 per cent company-profits error and the ~ 15 per cent incomes errors), the *I7/PR* index history (behind decaying RPI inflation), and the overseas rate-of-return normalisers (behind the current-account error) are unrecoverable from public sources. These errors are gated against regression, not chased to zero, because closing them would mean fitting to the answer.
4. **Dead and exogenised channels.** Exports, imports, CPI, the labour-demand block, and wages are exogenous in this configuration—held at the OBR path. Reforms whose first-round incidence runs through those channels (e.g. trade or supply-side labour measures) are outside the emulator’s current scope, and employment “accuracy” is a trivial identity over two passthrough inputs.
5. **Bridge assumptions are crude and declared.** The static costing lands entirely on nominal household disposable income, flat within each year; the economy’s feedback onto disposable income is not recycled; incidence is demand-side only; and corporation-tax reforms are refused rather than mistranslated. Supply-side responses belong to the suite’s OLG member.
6. **Vintage sensitivity and judgement contamination.** The ONS input snapshot is current-vintage data merged into an autumn-2025 baseline, so revision slack silently enters the add-factors; and the held-add-factor base window extends into quarters that are the OBR’s own forecast, so “held judgement” partly means “agree with the OBR.” The forecast scorecard also inherits a jump-off flatter from EFO seeding.
7. **Stabilised, not structural, investment closure.** The corporation-tax investment closure freezes market-sector output feedback at a tracking-pass reference to suppress an explosive accelerator loop; the mandated sign and boundedness invariants hold, but the closure is a stop-gap pending a populated market-sector supply block, and it is computationally expensive on the hosted service.

8. **No official counterpart to check dynamics against.** The OBR publishes the model listing and a description ([Office for Budget Responsibility, 2013](#)) but no official working paper, ready-reckoner of model multipliers, or reference simulation set for the current vintage; the emulator’s shock responses can be checked for sign, magnitude class, and convention-consistency, but not line-by-line against an official impulse response.

7 Conclusion

This paper has described an open Python emulator of the OBR’s macroeconometric model: 372 published EViews equations transpiled and solved in the open, anchored to the November 2025 EFO by the OBR’s own add-factor device, and connected to household-resolution microsimulation through a static-costing bridge that reproduces the official “static costing in, second-round effects out” workflow. The anchored baseline reproduces the published forecast to 0.18 per cent on GDP and is hard-gated in CI; the free-running model is documented as weak, with every residual error traced to a fixed bug or a floor imposed by unpublished OBR inputs. A worked 1p basic-rate reform yields a costing within the range of HMRC’s published ready reckoner, with small negative second-round GDP effects of the sign and size the official conventions imply.

The broader aim is methodological. Official fiscal scoring in the UK runs through a model that is described but not runnable by the public. Emulating it openly—with its judgement device made explicit, its passthroughs labelled, and its calibration failures published alongside its successes—moves the transparency frontier: researchers can now inspect the marginal propensities and error-correction speeds that shape UK costings, run counterfactual shocks through the official transmission mechanisms, and see precisely where the public information runs out. Within the PolicyEngine Macro suite, the emulator supplies the near-term, official-convention answer alongside the OLG model’s long-run structural answer and the microsimulation’s statutory one; scoring the same reform through all three, on a declared verification gradient, is the point.

Future work includes populating the market-sector supply block (retiring the stabilised investment closure), reviving the trade and labour-demand channels currently held exogenous, recycling income feedback through the bridge, and re-anchoring to each new EFO vintage as it is published.

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